Vuk Luzanin

Curriculum Vitae

Department of Computer Science
University of Texas at Austin

Research Interests

- Power, energy, and thermal efficiency of cloud and datacenter environments.
- Hardware-software co-design and computer architecture (RISC-V, AI accelerators) for power-efficient AI workloads.
- Parallel systems and high-performance computing for scalable, efficient AI workloads.

Education

2025–Present **PhD Computer Science**, *University of Texas at Austin*
Advisor: Prof. Jovan Stojkovic
Topic: Hardware–Software Co-design for Power-Efficient AI in Datacenters

2022–2026 **Bachelor of Science, Software Engineering**, *School of Electrical Engineering*
Thesis: Parallelization of the Feynman–Kac Algorithm on a Multicore Processor and Analysis of Real-World Applications **GPA:** 9.47/10

Research Experience

Aug 2026 - Present **Research Assistant, UT Austin**, *Advisor: Prof. Jovan Stojkovic*
Designing hardware and software to improve the power, energy, and thermal efficiency of scalable AI workloads in cloud and datacenter environments.

Apr 2025 - July 2026 **Research Student, Center for Data Processing, School of Electrical Engineering, University of Belgrade**, *Advisor: Prof. Marko Misic*
Parallel Feynman–Kac Algorithm: Performance Comparison and Real-World Applications. Compared parallel implementations of the algorithm using OpenMP, pthreads, and C++ threads on multicore CPUs against a CUDA implementation on GPUs, analyzing speedup and scalability. Evaluated how the resulting acceleration benefits real-world applications that rely on the algorithm.

Teaching Experience

Spring 2026 **Undergrad TA**, *University of Belgrade*, Computer Architecture, Algorithms and Data Structures, Computer Networks

Fall 2025 **Undergrad TA**, *University of Belgrade*, Computer Architecture, Algorithms and Data Structures, Object-oriented Programming, Fundamentals of Databases, Intelligent Systems, Introduction to Programming

Spring 2025 **Undergrad TA**, *University of Belgrade*, Computer Architecture, Algorithms and Data Structures, Computer Networks

Fall 2024 **Undergrad TA**, *University of Belgrade*, Computer Architecture, Algorithms and Data Structures, Object-oriented Programming, Fundamentals of Databases, Intelligent Systems, Introduction to Programming

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Industry Experience

Sept 2025 - Nov 2025 **Software Engineer Intern – AI Performance & Debugging Tools, Tenstorrent Inc.**, *Belgrade, Serbia*
Worked on a custom debugger for Tenstorrent's RISC-V–based AI accelerators in the Low Level Kernels, Debug team. Developed a new method for precisely reading a larger number of debug bus signals atomically at a single point in time, and interpreted them to expose characteristic hang signatures in multiprocessor workloads.

June 2025 - Aug 2025 **Software Engineer Intern, BlackRock**, *London and Belgrade*
Designed and implemented a centralized request routing microservice for the Private Markets division, serving as a unified entry point to internal BlackRock services. It discovers endpoints, finds routes from predefined configurations, and forwards requests via gRPC.

Oct 2024 - Dec 2025 **Embedded Software Engineer, Road Arrow – Formula Student Team**, *Belgrade, Serbia*
Developed embedded software for vehicle control systems on STM32 microcontrollers, covering real-time data acquisition, sensor integration, and communication protocols (CAN, I2C, UART, SPI). Implemented a CAN-based bootloader for remote code deployment and designed a new modular ECU, helping the team qualify for international competitions in Italy, Poland, and Croatia.

Scholarships & Awards

2023 - 2026 **Scholarship for Exceptionally Gifted Students**, Ministry of Education of the Republic of Serbia

2025 **3rd Place, Rivian Computer Vision Hackathon**

2022 **2nd & 3rd Place, Serbian National Mathematics Competition**

2022 **3rd Place, Serbian National Mathematics Competition**, Center for Talented Youth

2022 **3rd Place, Serbian National Physics Competition**

Selected Projects — available on GitHub

1. **Power-Efficient Orchestration and Cluster Reconfiguration for AI Workloads in Datacenters**
   Exploits workload flexibility to optimize request placement and hardware configuration while preserving performance and output quality. Develops a proactive orchestration algorithm that keeps each server within its provisioned power budget, together with a periodic cluster reconfiguration algorithm that adapts model placement, model precision, and hardware settings to changing workload demands.
2. **RAID Disk Management and Fault Tolerance System**
   Integrates RAID support into the xv6 (Unix-like) operating system, enabling configurations such as RAID0, RAID1, RAID0+1, RAID4, and RAID5. Focuses on fault tolerance, data redundancy, and safe parallel access for improved disk reliability and system performance, with optimizations for parallel execution of read and write requests in multiprocessor systems.

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3. **FPGA-based Ray Tracing Engine with Custom picoRISC CPU**
   Built from scratch a system that runs 3D ray tracing graphics on a Cyclone III FPGA. Designed a custom picoRISC CPU based on RISC-V that executes assembler code with a custom instruction set, a GPU for VGA output with its own SDRAM communication module, and a PS/2 keyboard controller.
4. **Operating System for RISC-V Architecture**
   Implemented a kernel from scratch with preemptive multi-threading for a RISC-V CPU with time sharing on a single CPU, supporting both synchronous and asynchronous context switching, event and signal handling, and semaphores, using C++ and GNU Assembler.
5. **Migration of Student Formula ECU Software to FreeRTOS Architecture**
   Migrated the student formula car's Engine Control Unit (ECU) software to FreeRTOS, introducing task-based scheduling, inter-task communication (queues, semaphores), and a more modular, responsive architecture for reliable real-time control of engine parameters.
6. **Bootloader for STM32 Microcontrollers**
   Developed a bootloader for multiple STM32 microcontrollers on a shared CAN bus using the OpenBLT library. A Raspberry Pi fetches precompiled firmware from GitLab and distributes it over CAN. When a microcontroller receives a request for a new version of its own code, it returns to bootloader mode to update, and then launches the user application.

Technologies

Languages: C, C++, CUDA, OpenMP, Python, Java, Spring, SQL, RISC-V Assembly, gRPC
Software Stack: *LLM serving* (vLLM), *Agent frameworks* (mini-SWE-agent), *Container environments* (Docker), *storage systems* (MongoDB)
Hardware: CPUs, GPUs, FPGAs, STM32 microcontroller, Raspberry Pi
Benchmarks: SWE-Bench, DABStep, Terminal-Bench
Languages: English – Advanced (TOEFL iBT 101/120, issued Nov 2025), German – Intermediate, Serbian – Native